



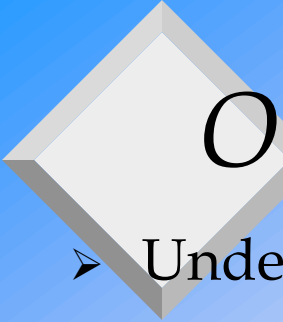
Architectural Components of DWH





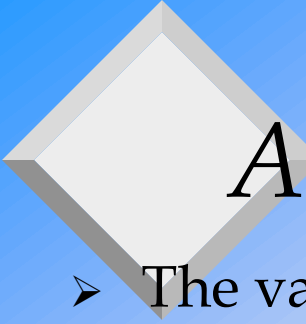
What We Learned in Last Lecture

- Overview of the Components
- Source Data Component
- Data Staging Component
- Data Storage Component
- Information Delivery Component
- Metadata Component
- Types of Metadata



Objectives of Today's Lecture

- Understand data warehouse architecture
- Learn about the architectural components
- Review the distinguishing characteristics of data warehouse architecture
- Examine how the architectural components supports the flow of data
- Comprehend what technical architecture means
- Study the functions and services of the architectural components



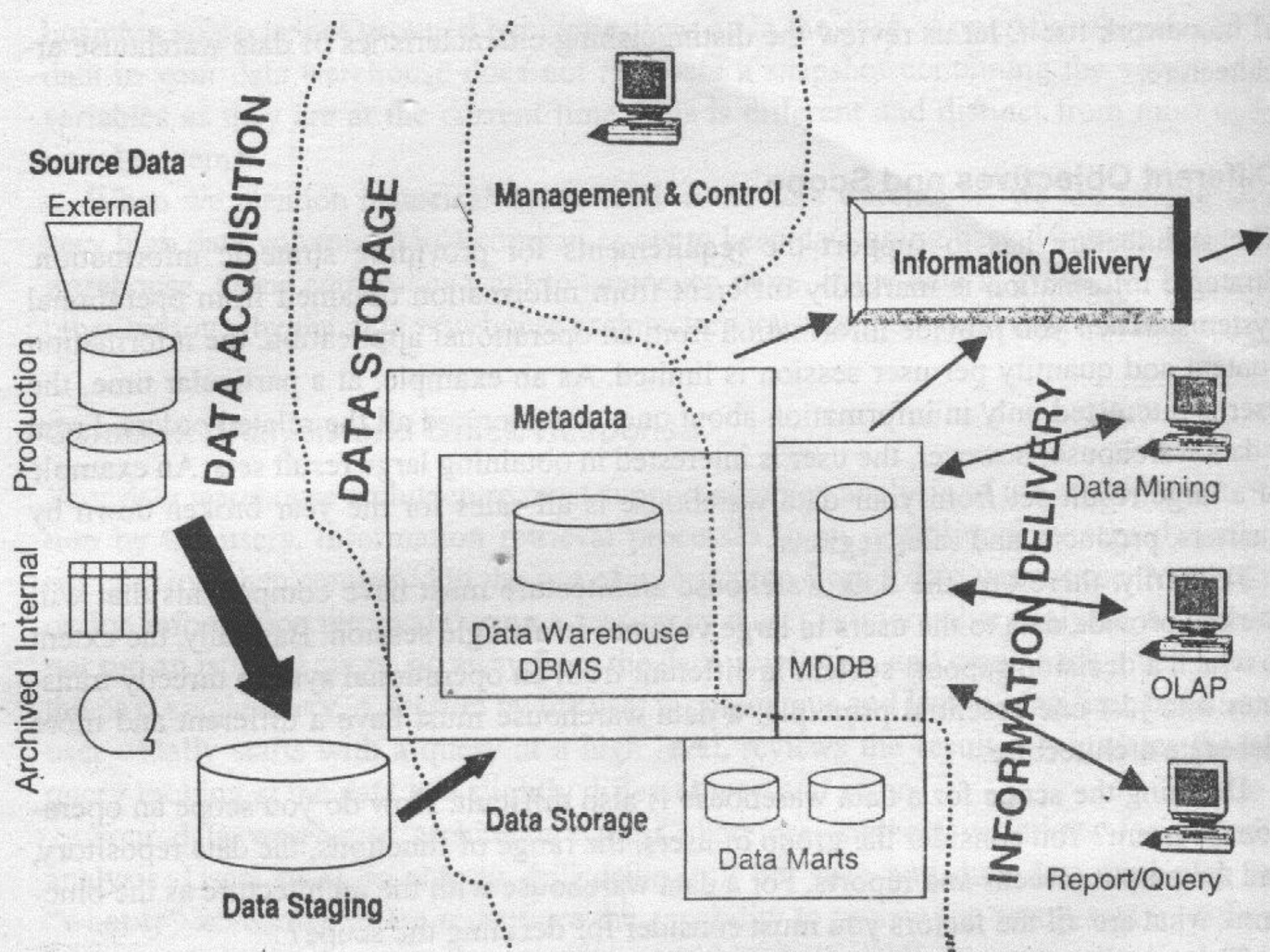
Architecture

- The various components of DWH together form the architecture of DWH
- The correct architecture is critical for the success of your DWH
- The architecture is further composed of the rules, procedures, and functions that enable your DWH to work and fulfill the business requirements
- Architecture is made up of technology that empowers your DWH
- It is a comprehensive blue print.
- The architecture defines the standards, measurements, general design and support techniques



Architecture contd..

- Architecture in three major areas
 - Data acquisition
 - Data storage
 - Information delivery
- We will concentrate on functions, services, and technical architecture in these major areas





Distinguishing Characteristics

- OLTP architecture is created by keeping in mind the context of operational systems
- DWH architecture is complex, wide and expensive



Different objectives and scope

- Architecture has to support the requirements for providing strategic information
- User is interested in obtaining large result sets
- Defining the scope of the DWH is also difficult
 - Have to configure the number of data sources
 - What are the external sources
 - Are you planning to include the private data too
 - Data granularity is also kept in mind



Data Content

- “Read only” data acts as primary component of the architecture
- Your DWH must support the sorting of data grouped by business “subjects”
- Architecture must support high volume of data because..
 - When we mention historical data stored in DWH, we are talking about very high data volumes.



Complex Analysis and Quick Response

- Your DWH architecture must support variations for providing analysis.
- Users must be able to..
 - Drill down
 - Roll up
 - Slice and dice data
 - Play with “what if” scenario
- User must have the capability to review the results in different output options like
 - Text
 - Tabular format
 - Graphical charts
- Your DWH architecture must have capability of quick response



Flexible and Dynamic

- After even implementation of DWH your DWH architecture must be flexible and Dynamic because
 - What if something was missed during development
 - User claims to add another dimension in the DWH
 - Business conditions changed
 - Business change calls for additional business requirement to be included in DWH



Meta data Driven

- Architecture should be meta data driven because..
 - Meta data surrounds the entire movement of data from source to the end users
 - It holds data about every phase of the movement, and, in a true sense, make the movement happen
- Data dictionary of DBMS of the operational system is just a faint shadow of the meta data in DWH



ARCHITECTURAL FRAMEWORK

- In data accusation, data storage, and information delivery areas, the architectural components serve specific purpose.
- Some of the architectural components governs the flow of data from beginning to end
- Some components perform standardizing, summarizing, and cleansing of data



ARCHITECTURAL FRAMEWORK

- At the data source level
 - Data staging architectural component governs the transformation, cleansing, and integration of data
- In DWH repository
 - Meta data architectural component is also a storage mechanism to contain data about data at every point of the flow of data from beginning to end
- At user end
 - The information delivery architectural component includes dependent data marts, special multi dimensional databases, and full range of query and reports facilities



ARCHITECTURAL FRAMEWORK

- The management and control module
 - It controls the entire DWH environment
 - It has two major functions
 - Constantly monitor all on going operations
 - Recover from problems when things go wrong
 - It manages and controls the data accusation functions
 - Ensures that the extracts and transformations are carried out correctly and in timely manners
 - Manages baking up significant parts of DWH and recovering from failure
 - Monitoring the growth and periodically archiving data from data warehouse
 - Governs data security
 - Provides authorized access to DWH
 - Interfaces end user information delivery component to ensure that information delivery is carried out properly

- ◆ This function is time-consuming
- ◆ Initial load moves very large volumes of data
- ◆ The business conditions determine the refresh cycles

**Data
Sources**

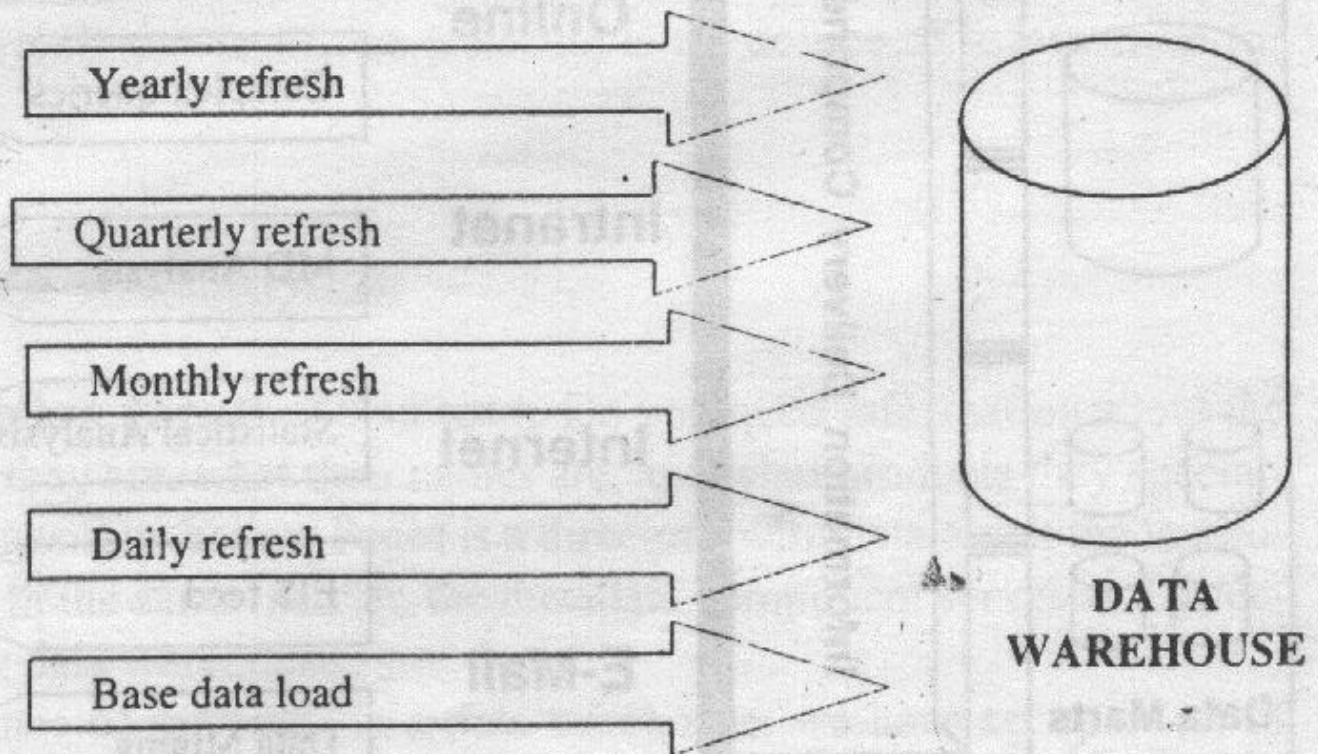
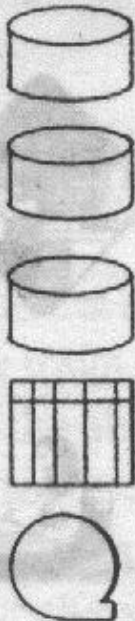
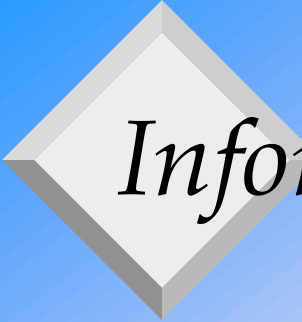


Figure 2-7 Data movements to the data warehouse.



Information Delivery Component

- Information is required for the wide community
 - Novice users
 - Casual users
 - Business analyst
 - Power users



Thank You Very Much